

Electrostatics

CHAPTER:- Electrostatics

COMPONENT/PARAMETER THEOREM/PRINCIPLE/CONDITION	FORMULA(S)	SI UNIT(S)
1) Electric Charge Q or q	$q = ne = it = \frac{V}{r} \cdot t = \frac{w}{v} \text{ OR } \vec{F} = q\vec{E}$	C
2) Electric Field E	$E = q.V = \frac{d\phi}{dA} = \frac{Kq}{r^2} = \frac{\sigma}{\epsilon_0}$	N/C
3) Linear Charge Density	$\lambda = \frac{Q}{L} = \frac{dQ}{dl}$	C/m
4) Surface Charge Density	$\sigma = \frac{Q}{A} = \frac{dQ}{dA}$	C/m ²
5) Volume Charge Density	$\rho = \frac{Q}{V} = \frac{dQ}{dV}$	C/m ³
6) Electric Flux	$\phi = \int E \cdot da = \vec{E} \cdot \vec{A} = E \cdot A \cos\theta$	Nm ² /C
7) Gauss Theorem	$\phi = \int E \cdot da = q/\epsilon_0$	Nm ² /C
8) Electric Force	$F = \frac{q1 \cdot q2}{4\pi\epsilon_0 r^2}$	N
9) Torque	$\tau = p \times E = p \cdot E \sin\theta$	Nm
10) Electric Dipole Moment	$p = q \cdot 2l = q \cdot d$	C-m
11) Acceleration of charged particles	$a = \frac{F}{m} = \frac{qE}{m}$	m/s ²
12) Electric Field due to Axial Line	$E_{axial} = \frac{2\vec{P}}{4\pi\epsilon_0 x^3}$	N/C
13)) Electric Field due to Equitorial Line	$E_{equitorial} = \frac{\vec{P}}{4\pi\epsilon_0 (x^2 + a^2)^{\frac{3}{2}}}$	N/C